ORIE 5135 Project

Peter Ye

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1 Question 1a: Natural ILP Formulation

Sets and Parameters:

- Set of exams: $V = \{1, 2, \ldots, m\}$
- Set of available time slots: $T = \{1, 2, \ldots, n\}$
- Number of proctors required by exam i: r_i
- Maximum proctor capacity per time slot: R
- Conflict edge set: $(i, i') \in E$ indicates exams i and i' share at least one student

Decision Variables:

- $x_{ij} \in \{0, 1\}$: equals 1 if exam i is assigned to time slot j, 0 otherwise
- $y_j \in \{0, 1\}$: equals 1 if time slot j is used, 0 otherwise

Model:

$$\begin{aligned}
\min \quad & z = \sum_{j \in T} y_j && (1)\\
\text{s.t.} \quad & \sum_{j \in T} x_{ij} = 1, \quad \forall\, i \in V && (2)\\
& \sum_{i \in V} r_i\, x_{ij} \le R\, y_j, \quad \forall\, j \in T && (3)\\
& x_{ij} + x_{i'j} \le y_j, \quad \forall\, (i, i') \in E,\ \forall\, j \in T && (4)\\
& y_j \ge y_{j+1}, \quad \forall\, j = 1, \ldots, n-1 && (5)\\
& x_{ij},\, y_j \in \{0, 1\}, \quad \forall\, i \in V,\ j \in T && (6)
\end{aligned}$$

The last constraint is a symmetry-breaking inequality. Since the time slots are interchangeable, any feasible schedule can be relabeled so that used slots come before unused ones; this constraint enforces that canonical labeling, removing a large amount of solution symmetry without changing the optimal objective value.

2 Question 1b: Clique-Strengthened ILP Formulation

The sets, parameters, decision variables, and all other constraints (including the symmetry-breaking inequality) remain the same as in Formulation 1a. The pairwise conflict constraints are replaced by clique inequalities.

Let $\mathcal{L}$ denote the set of all maximal cliques in the conflict graph G.

Clique Inequality:

$$\sum_{i \in C} x_{ij} \leq y_j, \quad \forall\, C \in \mathcal{L},\, \forall\, j \in T$$

Validity and dominance. Any two exams in a clique C pairwise share at least one student, so at most one exam from C can be assigned to time slot j; if any exam in C is in slot j then that slot is used, which gives $y_j = 1$. Hence $\sum_{i \in C} x_{ij} \leq y_j$. Each such clique inequality dominates the $\binom{|C|}{2}$ pairwise edge inequalities $x_{ij} + x_{i'j} \leq y_j$ for $i, i' \in C$: summing the pairwise inequalities only yields $\sum_{i \in C} x_{ij} \leq \frac{|C|}{2}\, y_j$, a strictly weaker bound for $|C| \geq 3$. Replacing pairwise edge constraints with clique inequalities therefore gives an LP relaxation that is at least as tight, and strictly tighter whenever the LP solution would otherwise place fractional mass on three or more vertices of a clique inside the same slot.

3 Question 2a: Extended Formulation

Let φ denote the collection of all feasible groupings. A subset $S \subseteq \{1, 2, \ldots, m\}$ belongs to φ if and only if:

1. **Independent set:** No two exams in S share a student (i.e., S is a stable set in G).
2. **Capacity:** $\sum_{i \in S} r_i \leq R$.

Decision Variable:

- $\lambda_S \in \{0, 1\}$: equals 1 if grouping S is selected, 0 otherwise

Model:

$$\min \quad z = \sum_{S \in \varphi} \lambda_S \tag{7}$$

$$\text{s.t.} \quad \sum_{\substack{S \in \varphi \\ i \in S}} \lambda_S = 1, \quad \forall\, i \in \{1, 2, \ldots, m\} \tag{8}$$

$$\lambda_S \in \{0, 1\}, \quad \forall\, S \in \varphi \tag{9}$$

This is a set-partitioning model with one variable per feasible grouping, which is exactly what we implement and use as the baseline in Question 3. We assume that the number of available time slots n is large enough to accommodate the optimal schedule, so no explicit upper bound on the number of selected groupings is needed (this holds for every instance in the two datasets, where $n = m$).

4 Question 2b: Set Covering vs. Set Partitioning

In the set partitioning formulation (used in Question 2a), each exam must belong to *exactly one* selected grouping ($= 1$). This requires enumerating *all* feasible groupings, including non-maximal ones, because restricting to only maximal groupings may make it impossible to partition the exams such that each appears in exactly one group.

An alternative is the set covering formulation, where each exam must belong to *at least one* selected grouping (≥ 1). Since the objective minimizes the number of selected groupings, no optimal solution will redundantly cover an exam in multiple groupings.

The set covering formulation allows us to restrict the variables to only **maximal** feasible groupings — groupings to which no additional exam can be added without violating either the proctor capacity or a conflict constraint. This significantly reduces the number of variables.

Despite this difference, the two formulations are equivalent: any solution of the set partitioning model can be converted to a solution of the set covering model with the same objective value (by expanding each non-maximal grouping to a maximal one), and any solution of the set covering model can be converted back to a set partitioning solution with the same value, by shrinking each maximal grouping to a subset that ensures every exam is contained in exactly one of them. Their LP relaxation values also coincide.

In practice, the set covering formulation with maximal groupings is preferable because:

1. The number of variables is smaller (though still exponential in the worst case).
2. The LP relaxation values are identical.
3. LPs with inequality constraints ($\geq$) are generally easier for solvers than LPs with equality constraints ($=$).

The set covering formulation replaces the equality constraints with inequalities:

$$\min \quad z = \sum_{S \in \mathcal{S}} \lambda_S \tag{10}$$

$$\text{s.t.} \quad \sum_{\substack{S \in \mathcal{S} \\ i \in S}} \lambda_S \geq 1, \quad \forall\, i \in \{1, 2, \ldots, m\} \tag{11}$$

$$\lambda_S \in \{0, 1\}, \quad \forall\, S \in \mathcal{S} \tag{12}$$

where $\mathcal{S}$ denotes the collection of all **maximal** feasible groupings. A feasible grouping S is maximal if no additional exam can be added to S without violating either the proctor capacity constraint or the independence requirement.

We did not implement the set covering version, but compared with set partitioning we would expect it to be faster on both ends of the pipeline. Enumeration would produce fewer columns, since only maximal groupings need to be stored rather than every smaller subset visited along the way. Each LP solve would also be slightly cheaper, because there are fewer variables and inequality constraints are usually easier for the solver to handle than equalities. The root LP value stays the same, so the bound quality is unchanged.

5 Question 3: Computational Experiments

All experiments were run with Gurobi (via `gurobipy`) on a single machine. The implementation reads each instance directly from its JSON file and builds the three formulations described in Sections 1 and 2. For every formulation, we record (i) the LP relaxation value at the root, (ii) the number of branch-and-bound nodes explored, and (iii) the wall-clock solve time — as reported by Gurobi for Formulations 1a and 1b, and as total wall time including the enumeration of feasible groupings for Formulation 2. Each instance has $m = n$ and $R = 25$, with conflict-graph density close to 0.5.

5.1 Question 3(a): Dataset 1 (time limit 120 s)

Dataset 1 contains ten instances of growing size, ranging from $m = 35$ ($|E| = 292$) to $m = 75$ ($|E| \approx 1400$). Table 1 reports the results for the three formulations on every instance.

(i) LP relaxation values. The natural formulation (1a) and the clique-strengthened formulation (1b) produce *identical* root LP values on every instance of Dataset 1 (they differ only at the last decimal due to floating-point noise). At first glance this is surprising, since the clique inequality strictly dominates the corresponding pairwise edge inequalities. The explanation has two parts. First, on these graphs with edge density $\sim$0.5, the LP optimum is largely determined by the proctor capacity bound $\sum_i r_i/R$ together with the symmetry-breaking constraints $y_j \geq y_{j+1}$; once the LP saturates these constraints, the additional clique inequalities are not active. Second, Gurobi's presolve very likely generates clique cuts from the pairwise edge constraints automatically, so writing the cliques explicitly does not add new information at the root LP. Formulation 2 is provably at least as tight as Formulations 1a and 1b, and we observe *strict* improvement only on the smallest, sparsest instances (01–03), where the conflict-graph structure dominates over the capacity bound; from instance 04 onward the capacity bound becomes binding and all three formulations report the same LP value.

Table 1: Dataset 1 results. "Opt?" indicates whether Gurobi proved optimality within the 120 s time limit. "LP" is the value of the LP relaxation at the root. For Formulation 2, "Time" includes the time spent enumerating feasible groupings.

Instance	(m, $\lvert E\rvert$)	Form.	LP	Obj.	Opt?	Nodes	Time (s)
01	(35, 292)	1a Natural	6.32	8	Y	1	1.20
		1b Clique	6.32	8	Y	1	2.35
		2 Extended	7.18	8	Y	1	0.04
02	(38, 345)	1a Natural	7.32	9	Y	943	1.89
		1b Clique	7.32	9	Y	791	7.94
		2 Extended	7.92	9	Y	1	0.08
03	(40, 369)	1a Natural	7.88	9	Y	209	2.69
		1b Clique	7.88	9	Y	153	3.24
		2 Extended	8.12	9	Y	1	0.10
04	(60, 886)	1a Natural	12.84	13	Y	207 271	99.36
		1b Clique	12.84	13	N	9 104	120.05
		2 Extended	12.84	13	Y	1	0.61
05	(63, 947)	1a Natural	12.40	13	Y	1	10.89
		1b Clique	12.40	13	Y	1	37.28
		2 Extended	12.40	13	Y	1	0.58
06	(65, 1057)	1a Natural	12.88	14	N	32 091	120.04
		1b Clique	12.88	14	N	3 781	120.03
		2 Extended	12.88	13	Y	413	2.44
07	(68, 1109)	1a Natural	12.68	13	Y	6 516	21.26
		1b Clique	12.68	13	Y	1 524	101.51
		2 Extended	12.68	13	Y	1	1.76
08	(75, 1380)	1a Natural	15.36	16	Y	1	20.02
		1b Clique	15.36	16	Y	1	115.48
		2 Extended	15.36	16	Y	1	0.77
09	(75, 1372)	1a Natural	14.16	15	Y	1	15.44
		1b Clique	14.16	75	N	1	120.02
		2 Extended	14.16	15	Y	1	1.13
10	(75, 1423)	1a Natural	14.68	16	N	29 094	120.01
		1b Clique	14.68	73	N	1	120.04
		2 Extended	14.68	15	Y	1	0.89

(ii) Branch-and-bound nodes vs. solve time. Equal LP relaxation values do *not* imply equal branch-and-bound effort. The clearest example is instance 04: all three formulations have LP value 12.84, but Formulation 2 closes the gap at the root (1 node, 0.6 s), Formulation 1b explores $\sim 9\,000$ nodes and times out, and Formulation 1a explores $\sim 2\times 10^5$ nodes (99 s). The same pattern appears on instances 02, 06, and 07. Two factors drive this. (a) The model size differs dramatically: 1b adds one constraint per (maximal clique, time slot) pair, which on dense graphs yields many thousands of additional rows, making each node LP slower to solve and the branching/heuristic stage less effective at finding incumbents. (b) The extended formulation has very few variables relative to its information content (each λ_S encodes an entire feasible slot), so Gurobi's primal heuristics typically recover an integer-optimal incumbent immediately at the root. Thus our data suggests that LP bound quality is necessary but not sufficient for fast B&B; the structural compactness of the formulation and the difficulty of each individual node LP matter just as much. A striking failure mode is visible on instances 09 and 10 of Formulation 1b, which return objective values of 75 and 73 respectively—near the trivial "one exam per slot" upper bound of $m = 75$—meaning that within the 120 s limit Gurobi's heuristics could not find a tight feasible solution under this formulation.

(iii) Scaling with instance size. The trend across the ten instances is clear. Formulation 2 (extended) scales the best *in solve time*: nine of the ten instances are closed with a single B&B node, and total time stays below 2.5 s. The bottleneck of Formulation 2 is purely combinatorial enumeration of the feasible groupings, which is still tractable for $m \leq 75$ at this density but grows exponentially. Formulations 1a and 1b both suffer rapidly increasing node counts as m and $|E|$ grow: the natural formulation times out on instances 06 and 10 and the clique formulation times out on instances 04, 06, 09, and 10. The clique formulation typically explores fewer nodes than the natural formulation when both finish (e.g., 1 524 vs. 6 516 on instance 07), but its per-node cost is much higher because of the larger constraint matrix, so its wall-clock time is consistently worse on this dataset. In summary, growth rates differ substantially: Formulation 2 grows in *enumeration* cost, while Formulations 1a and 1b grow in *B&B* cost, with 1b having the worst wall-clock growth in our experiments.

5.2 Question 3(b): Dataset 2 (time limit 600 s)

Dataset 2 contains nine larger instances with $m \in \{80, \ldots, 90\}$ and $|E|$ up to ~ 2000. Table 2 reports the results for Formulations 1a and 1b. Formulation 2 is omitted because the enumeration of feasible groupings is no longer practical at this scale (see discussion below).

Table 2: Dataset 2 results (time limit 600 s).

Instance	$(m, \|E\|)$	Form.	LP	Obj.	Opt?	Nodes	Time (s)
01	(80, 1584)	1a Natural	16.32	17	Y	1	22.26
		1b Clique	16.32	17	Y	1	205.09
02	(82, 1682)	1a Natural	16.08	17	Y	1	15.55
		1b Clique	16.08	17	Y	1	347.32
03	(83, 1726)	1a Natural	16.28	17	Y	1	32.54
		1b Clique	16.28	17	Y	1	355.03
04	(87, 1843)	1a Natural	16.76	17	Y	9 995	75.71
		1b Clique	16.76	17	Y	1	556.95
05	(87, 1867)	1a Natural	16.60	17	Y	2 962	64.76
		1b Clique	16.60	81	N	1	600.09
06	(87, 1821)	1a Natural	17.76	18	Y	3 434	51.99
		1b Clique	17.76	18	N	216	600.14
07	(82, 1660)	1a Natural	16.88	17	Y	20 031	97.49
		1b Clique	16.88	18	N	2 347	600.08
08	(85, 1805)	1a Natural	16.84	17	Y	37 245	214.19
		1b Clique	16.84	18	N	1	600.06
09	(90, 2032)	1a Natural	18.00	19	N	153 156	600.03
		1b Clique	18.00	20	N	1	600.12

Comparison and computational limits. The qualitative pattern observed on Dataset 1 is amplified here. The two LP relaxations remain identical on every instance, again because the capacity bound and the symmetry-breaking constraints dominate the structure of the LP optimum and Gurobi's presolve already exploits clique structure internally. In terms of solving the ILP, however, Formulation 1a clearly outperforms Formulation 1b: 1a closes eight of the nine instances within the time limit (only the largest, instance 09, times out), whereas 1b proves optimality on only *four* of the nine. On the remaining five instances, 1b often returns notably weaker incumbents (e.g., 81 on instance 05, 20 on instance 09 against 1a's 19), reflecting that the heavy constraint matrix slows both LP solves at each node and primal heuristics, so the solver cannot find tight integer solutions in time. This is the same trade-off seen on Dataset 1, scaled up: clique strengthening that should in principle help branching becomes counterproductive once the number of maximal cliques explodes on dense graphs of this size. Even the natural formulation 1a starts to scale poorly: solve time grows from ~ 20 s at $m = 80$ to over 200 s at $m = 85$, with 1.5×10^5 nodes explored on the largest instance before timing out. Both formulations are therefore approaching their practical limit at $m \approx 90$ on graphs of edge density ~ 0.5.

Why the extended formulation is not applicable. The extended formulation has one variable per feasible grouping, where a grouping is both an independent set in G and a knapsack-feasible subset under the proctor budget. On Dataset 1 (up to $m = 75$) the recursive enumeration in our implementation is still feasible. On Dataset 2 the situation changes qualitatively: with $m \approx 90$ and edge density ~ 0.5, the conflict graph contains many large independent sets, and the proctor capacity $R = 25$ is not tight enough to prune the search aggressively. The number of feasible groupings grows exponentially in m on graphs of this density, so on these instances it is already astronomically large; even storing the complete variable set in memory is impractical, let alone passing it to Gurobi. In other words, the extended formulation trades a polynomial-size model (Formulations 1a/1b) for an exponentially large one whose strength comes precisely from that exhaustive enumeration; once enumeration is no longer feasible, the formulation cannot be built explicitly, which is why Question 3(b) restricts the comparison to the two compact formulations.